Chapter 3: The Water Cycle

The water cycle describes how water moves continuously between the oceans, the atmosphere and the land. It is driven by energy from the Sun and by gravity. Understanding the water cycle is important because it controls the supply of fresh water, shapes weather and climate, and moves heat around the planet.

3.1 Evaporation and Transpiration

Evaporation is the process in which liquid water changes into water vapour when it absorbs heat energy. Most evaporation happens from the surface of the oceans, which cover about 71 percent of the Earth. Transpiration is the release of water vapour from the leaves of plants through small pores called stomata. Together, evaporation and transpiration are often called evapotranspiration.

The rate of evaporation increases when temperatures are high, when the air is dry and when wind removes moist air from the surface. For example, a puddle dries much faster on a hot, windy afternoon than on a cool, still morning.

3.2 Condensation and Cloud Formation

Condensation is the change of water vapour back into liquid water droplets when air cools. As warm, moist air rises, it expands and cools. When it cools to its dew point, the vapour condenses onto tiny particles of dust or salt, which are known as condensation nuclei. Billions of these droplets together form clouds.

- Cumulus clouds form when warm air rises quickly on sunny days.
- Stratus clouds form in flat layers when a large mass of air cools slowly.
- Cirrus clouds form high in the atmosphere and are made of ice crystals.

3.3 Precipitation

Precipitation is any form of water that falls from clouds to the ground, including rain, snow, sleet and hail. Precipitation happens when cloud droplets collide and grow until they are too heavy to stay in the air. The type of precipitation depends mainly on the temperature of the air between the cloud and the ground.

3.4 Collection, Infiltration and Runoff

After precipitation, water collects in oceans, lakes, rivers and underground stores. Infiltration is the movement of water from the surface into the soil. Water that soaks deep into the ground becomes groundwater, which is stored in porous rocks called aquifers. Surface runoff is water that flows over the land into streams and rivers when the ground cannot absorb it fast enough.

Human activity changes these processes. Cities covered in concrete and tarmac reduce infiltration and increase surface runoff, which raises the risk of flooding. Therefore, many cities now use green roofs and permeable paving to slow water down.

Summary

- The Sun's energy drives evaporation and transpiration.
- Cooling air causes condensation, which forms clouds.
- Precipitation returns water to the surface, where it is stored, infiltrates the ground or becomes runoff.
- The water cycle is a closed system: the total amount of water on Earth stays almost constant.